COURSE 3 · PHYSICS

Physics

Theory reader, audio edition

Motion, forces, energy, momentum, waves, electricity, magnetism, heat, and the modern physics of relativity, quanta and the nucleus.

9 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 3.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

3.1 Motion (kinematics)
3.2 Forces and Newton's laws
3.3 Energy, work and power
3.4 Momentum, collisions and rotation
3.5 Waves and sound
3.6 Electricity and circuits
3.7 Magnetism and electromagnetism
3.8 Thermodynamics and heat
3.9 Modern physics

Physics looks for the smallest set of rules that explain the widest range of things. This course moves from the motion of everyday objects to electricity and magnetism and on to the twentieth-century revolutions of relativity and quantum theory. It leans on Course 1 (vectors, trigonometry, units) and Course 2 (rearranging, graphs, a little calculus).

Module 3.1

Motion (kinematics)

Kinematics describes motion without asking what causes it. With four quantities, displacement, velocity, acceleration and time, and four equations linking them, you can predict where almost anything will be.

By the end of this module you can

- Distinguish distance, displacement, speed, velocity and acceleration
- Read motion graphs: gradients and areas
- Use the constant-acceleration equations, including for projectiles

Describing motion

Velocity is the rate of change of displacement: how fast, and in which direction. Acceleration is the rate of change of velocity. A car going round a roundabout at a steady speed is accelerating, because its direction keeps changing. Units matter: velocity in metres per second, acceleration in metres per second per second, written metres per second squared.

On a displacement-time graph, the gradient is velocity. On a velocity-time graph, the gradient is acceleration and the area underneath is displacement. These two facts, which are the derivative and the integral from Module 2.8, let you extract a great deal from a single graph.

The constant-acceleration equations

When acceleration is constant, four equations, sometimes called the SUVAT equations after their symbols, connect initial velocity u, final velocity v, acceleration a, time t and displacement s [Equation 3.1-1]. Each equation leaves out one of the five quantities. Pick the one that leaves out the quantity you neither know nor need.

Near Earth's surface, anything falling freely accelerates downwards at about nine point eight metres per second squared, whatever its mass, provided air resistance is small. Galileo is said to have tested this at Pisa; Apollo 15 astronauts confirmed it on the Moon with a hammer and a feather.

Projectiles

A thrown ball moves sideways and up-and-down at the same time, and the two motions are independent. Horizontally, with no air resistance, nothing pushes it, so its horizontal velocity stays constant. Vertically, it accelerates downwards at g. Resolve the launch velocity into components [Equation 3.1-2], treat each direction separately, and connect them through the shared time of flight. On level ground, the range is greatest for a launch at forty-five degrees [Equation 3.1-3].

Key ideas to carry forward

- Gradient of x–t is velocity; gradient of v–t is acceleration; area under v–t is displacement
- SUVAT equations need constant acceleration
- Projectiles: horizontal and vertical motions are independent

Module 3.2

Forces and Newton's laws

Newton's three laws explain why things move as they do. They are among the most successful ideas in science, good enough to land spacecraft on comets, and they all come down to one relationship: force changes motion.

By the end of this module you can

- State and apply Newton's three laws
- Draw free-body diagrams and resolve forces
- Work with weight, normal force, friction, tension and circular motion

The three laws

The first law: an object stays at rest or keeps moving at constant velocity unless a net force acts on it. This was a radical idea; everyday experience suggests things naturally slow down, but that is friction at work. The second law: net force equals mass times acceleration [Equation 3.2-1]. A bigger force gives a bigger acceleration; a bigger mass gives a smaller one. The third law: when A pushes on B, B pushes back on A with an equal and opposite force. These pairs act on different objects, so they never cancel each other.

Common forces

Weight is the gravitational pull on a mass: mass times g [Equation 3.2-2]. A two-kilogram bag weighs about twenty newtons. The normal force is the push of a surface, perpendicular to it. Friction opposes sliding and is roughly proportional to the normal force [Equation 3.2-3], with a coefficient that depends on the surfaces. Tension is the pull of a rope or cable.

A free-body diagram shows one object and every force acting on it as an arrow. It is the most important habit in mechanics. Draw it, choose axes, resolve forces into components, then apply the second law in each direction. On a slope, it pays to tilt your axes along and perpendicular to the surface: weight then splits into mg sine theta down the slope and mg cos theta into it.

Circular motion

An object moving in a circle at steady speed is accelerating towards the centre, because its direction is always changing. The centripetal acceleration is v squared over r [Equation 3.2-4], so the required inward force is m v squared over r. This force is not a new kind of force; it is whatever is supplying the pull: tension in a string, friction on car tyres, or gravity for an orbiting satellite.

Key ideas to carry forward

- Net force = mass × acceleration
- Action–reaction pairs act on different objects
- Free-body diagram first, then resolve and apply F = ma
- Circular motion needs an inward force mv^2/r

Module 3.3

Energy, work and power

Energy is the bookkeeping of physics. It comes in many forms, moves between them constantly, and is never created or destroyed. Tracking it often solves problems that forces alone make hard.

By the end of this module you can

- Calculate work done by a force
- Use kinetic and gravitational potential energy and conservation of energy
- Calculate power and efficiency

Work

In physics, work is done when a force moves something: force times distance moved in the direction of the force [Equation 3.3-1]. Holding a heavy bag still does no work in this sense, however tiring. Push at an angle and only the component along the motion counts, which is the dot product from Module 1.6. Work is measured in joules: one newton pushing through one metre is one joule.

Kinetic and potential energy

A moving mass has kinetic energy, one half m v squared [Equation 3.3-2]. Because of the square, doubling speed quadruples kinetic energy, which is why stopping distances grow so quickly with speed. A mass raised in a gravitational field stores potential energy, m g h [Equation 3.3-3]. A stretched spring stores elastic energy.

Conservation of energy says the total stays the same. A ball dropped from height h loses potential energy and gains an equal amount of kinetic energy, so its speed at the bottom is the square root of two g h, independent of mass. Friction and air resistance do not destroy energy; they convert it to heat, which is harder to use.

Power and efficiency

Power is the rate of doing work or transferring energy: energy divided by time [Equation 3.3-4], measured in watts. A one-kilowatt heater transfers a thousand joules every second. Electricity bills charge for kilowatt-hours, which is energy, not power: one kilowatt for one hour, three point six million joules.

Efficiency is useful energy out divided by total energy in [Equation 3.3-5]. No real machine reaches one hundred per cent. A car engine manages perhaps thirty per cent; an electric motor ninety. The rest goes as heat, which links forward to thermodynamics in Module 3.8.

Key ideas to carry forward

- Work = force × distance in the direction of the force
- KE = ½mv^2; GPE = mgh; total energy is conserved
- Power = energy ÷ time; a kWh is energy
- Efficiency = useful out ÷ total in

Module 3.4

Momentum, collisions and rotation

Momentum is mass in motion. It is conserved in every collision and explosion, which makes it the go-to tool when objects hit each other. Rotation has its own versions of force, mass and momentum, and they explain why spinning skaters speed up.

By the end of this module you can

- Use momentum and impulse, and conservation of momentum
- Distinguish elastic and inelastic collisions
- Describe torque, moment of inertia and angular momentum

Momentum and impulse

Momentum is mass times velocity [Equation 3.4-1], a vector. Newton actually wrote his second law as force equals the rate of change of momentum. Rearranged, force times time equals the change in momentum, a quantity called impulse [Equation 3.4-2]. This is why airbags and crumple zones save lives: they stretch the time of a collision, so the same change in momentum happens with a smaller force.

Conservation of momentum

If no external force acts, the total momentum of a system stays constant [Equation 3.4-3]. In a collision, whatever momentum one object loses, the other gains. A two-kilogram ball at three metres per second hitting a stationary four-kilogram ball and sticking to it: total momentum six kilogram metres per second before, so six afterwards, shared by six kilograms, giving one metre per second.

Kinetic energy is conserved only in perfectly elastic collisions, like those between gas molecules. In everyday collisions some kinetic energy becomes heat, sound and deformation; these are inelastic. Momentum is conserved either way. Rockets work by conservation of momentum, throwing exhaust backwards to move forwards, even in empty space.

Rotation

Turning effects depend on where a force is applied. Torque, or moment, is force times perpendicular distance from the pivot [Equation 3.4-4]. That is why door handles are far from the hinges. An object is balanced when clockwise and anticlockwise moments are equal.

Rotating objects resist changes to their spin according to their moment of inertia, which depends on how far the mass sits from the axis. Angular momentum, moment of inertia times angular velocity [Equation 3.4-5], is conserved when no external torque acts. A skater pulling in her arms reduces her moment of inertia, so she spins faster. Collapsing stars do the same, which is why neutron stars, met in Course 5, can spin hundreds of times a second.

Key ideas to carry forward

- Momentum $p = mv$; impulse $F\Delta t = \Delta p$
- Total momentum is conserved without external forces
- Elastic collisions conserve KE; inelastic do not
- Torque = force × perpendicular distance; angular momentum is conserved

Module 3.5

Waves and sound

A wave carries energy from place to place without carrying matter. Ripples, sound, earthquakes and light are all waves, and they share a small set of properties and one master equation.

By the end of this module you can

- Distinguish transverse and longitudinal waves and their properties
- Use the wave equation v = fλ
- Explain superposition, interference, standing waves, resonance and the Doppler effect

What a wave is

In a transverse wave, like a ripple on a rope, the medium moves at right angles to the direction the wave travels. In a longitudinal wave, like sound, the medium moves back and forth along the direction of travel, forming compressions and rarefactions. Every wave has an amplitude, a wavelength, the distance between peaks, a frequency, the number of waves per second, and a period, the time for one wave.

The wave equation

Wave speed equals frequency times wavelength [Equation 3.5-1]. Sound in air travels at about three hundred and forty metres per second, so a middle C at two hundred and sixty-two hertz has a wavelength of about one point three metres. The speed is set by the medium, not the source: sound travels faster in water and faster still in steel. When a wave enters a new medium its frequency stays the same and its wavelength changes.

Superposition and interference

When waves meet, they add: this is superposition. Peaks meeting peaks give a bigger wave, constructive interference. Peaks meeting troughs cancel, destructive interference, which is how noise-cancelling headphones work. A wave reflected back on itself can form a standing wave, with fixed nodes that never move. Guitar strings and organ pipes vibrate in standing-wave patterns, and their allowed wavelengths set the notes [Equation 3.5-2].

Every object has natural frequencies. Push it at one of them and the amplitude builds up: resonance. It shatters wine glasses, tunes radios and must be designed out of bridges.

The Doppler effect

A siren sounds higher as it approaches and lower as it recedes. Waves from a moving source bunch up ahead and stretch out behind [Equation 3.5-3]. The same effect for light, a shift towards red for receding sources, is how astronomers discovered that the universe is expanding, which you will meet in Course 5.

Key ideas to carry forward

- Transverse: perpendicular oscillation; longitudinal: parallel
- $v = f\lambda$; speed is set by the medium
- Waves superpose; standing waves have fixed nodes
- Doppler: approaching source, higher frequency

Module 3.6

Electricity and circuits

Electricity is the flow of charge. Three quantities, voltage, current and resistance, linked by one simple law, explain everything from a torch to the wiring in your house.

By the end of this module you can

- Define charge, current, potential difference and resistance
- Use Ohm's law and the rules for series and parallel circuits
- Calculate electrical power and energy

Charge, current and voltage

Electric charge comes in two kinds, positive and negative; like charges repel and unlike attract. Charge is measured in coulombs. Current is the rate of flow of charge, coulombs per second, called amperes [Equation 3.6-1]. In metal wires, the moving charges are electrons, drifting surprisingly slowly, a fraction of a millimetre per second, though the signal travels near the speed of light.

Potential difference, or voltage, is the energy given to or taken from each coulomb of charge [Equation 3.6-2]. A nine-volt battery gives every coulomb nine joules. A useful picture is water in pipes: voltage is like pressure, current is like flow rate, and resistance is like narrowness of the pipe.

Ohm's law and resistance

For many conductors, current is proportional to voltage: voltage equals current times resistance [Equation 3.6-3]. Resistance, measured in ohms, depends on the material, the length and the thickness of the conductor. Longer and thinner wires resist more.

Series and parallel

In series, components are in a single loop. The same current flows through each, voltages add up, and resistances add [Equation 3.6-4]. In parallel, components sit on separate branches. Each gets the full voltage, currents add up, and the combined resistance is less than any single branch [Equation 3.6-5]. Household appliances are wired in parallel, so each gets mains voltage and can be switched independently.

Power

Electrical power is voltage times current [Equation 3.6-6]. A two-thousand-watt kettle on two hundred and thirty volts draws about nine amps. Combining with Ohm's law gives two other forms, I squared R and V squared over R, which explain why power lines run at high voltage: for the same power, a higher voltage means less current, and the heating loss, I squared R, drops dramatically.

Key ideas to carry forward

- Current = charge per second; voltage = energy per coulomb
- $V = IR$
- Series: R adds, current same. Parallel: 1/R adds, voltage same
- $P = VI = I^2R = V^2/R$

Module 3.7

Magnetism and electromagnetism

Electricity and magnetism turn out to be two faces of one force. Moving charges make magnetic fields, and changing magnetic fields drive currents. Those two facts power every motor and generator, and lead to the nature of light itself.

By the end of this module you can

- Describe magnetic fields of magnets and currents
- Calculate the force on a current or moving charge in a magnetic field
- Explain electromagnetic induction, generators and transformers

Magnetic fields

A magnet has north and south poles; like poles repel. The space around it holds a magnetic field, drawn as lines running from north to south. Earth itself is a magnet, which is why compasses work. In 1820 Hans Christian Ørsted noticed that a current-carrying wire deflected a compass needle: electric currents make magnetic fields. Around a straight wire, the field lines are circles, and the right-hand grip rule gives their direction.

The motor effect

A current-carrying wire in a magnetic field feels a force, perpendicular to both the current and the field [Equation 3.7-1]. A single moving charge feels a similar force [Equation 3.7-2]. This is the principle of the electric motor: a coil carrying current in a magnetic field experiences forces that make it turn. The force on a moving charge also explains why charged particles from the Sun spiral along Earth's field lines into the polar regions, producing the aurora.

Induction

In 1831 Michael Faraday found the reverse effect: a changing magnetic field through a coil induces a voltage in it [Equation 3.7-3]. Move a magnet into a coil and a current flows; hold it still and the current stops. The faster the change, the bigger the voltage. Generators spin coils in magnetic fields to produce alternating current, and almost all of the world's electricity is made this way.

Transformers use induction to change voltage. An alternating current in one coil makes a changing field that induces a voltage in a second coil. The ratio of the voltages equals the ratio of the turns [Equation 3.7-4]. This is how power stations step voltage up for transmission and substations step it down for homes.

James Clerk Maxwell combined these laws in the 1860s and found that changing electric and magnetic fields can sustain each other as a wave travelling at three hundred thousand kilometres per second: the speed of light. Light, he concluded, is an electromagnetic wave.

Key ideas to carry forward

- Currents create magnetic fields
- Force on a current in a field: F = BIL (perpendicular)
- Changing magnetic flux induces a voltage
- Transformers: Vs/Vp = Ns/Np; light is an electromagnetic wave

Module 3.8

Thermodynamics and heat

Thermodynamics is the physics of heat, temperature and energy conversion. Its laws are among the most general in science: they constrain engines, refrigerators, living cells and the fate of the universe.

By the end of this module you can

- Distinguish heat, temperature and internal energy; use the kelvin scale
- Calculate energy for temperature change and change of state
- State the laws of thermodynamics and explain entropy and engine efficiency

Temperature and heat

Temperature measures the average kinetic energy of the particles in a substance. Heat is energy transferred because of a temperature difference, always flowing from hot to cold. Scientists use the kelvin scale, which starts at absolute zero, the temperature at which particle motion is at a minimum [Equation 3.8-1]. A change of one kelvin equals a change of one degree Celsius; zero degrees Celsius is two hundred and seventy-three kelvin.

Heating and changing state

The energy needed to warm a substance is mass times specific heat capacity times temperature change [Equation 3.8-2]. Water's specific heat is unusually high, about four thousand two hundred joules per kilogram per kelvin, which is why oceans moderate climate. During melting or boiling, energy goes into breaking bonds between particles rather than raising temperature. This latent heat [Equation 3.8-3] explains the flat sections on a heating curve and why steam burns are worse than boiling-water burns.

The laws of thermodynamics

The zeroth law says objects in contact reach the same temperature. The first law is conservation of energy: the change in internal energy equals heat added minus work done by the system [Equation 3.8-4]. The second law is the deep one: heat flows spontaneously from hot to cold, never the reverse, and the total entropy of an isolated system never decreases. Entropy is, roughly, a measure of how spread out or disordered energy is.

The second law limits every heat engine. Some heat must always be dumped into a cold reservoir, so no engine can turn heat completely into work. The best possible efficiency depends only on the hot and cold temperatures in kelvin [Equation 3.8-5]. A power station with steam at eight hundred kelvin and cooling at three hundred can never beat about sixty-two per cent. The third law adds that absolute zero can be approached but never reached.

Bridge from the social sciences

The second law is often invoked loosely in social commentary. In physics it is precise: it concerns energy and microscopic arrangements, and it says nothing against local order such as living things, which export entropy to their surroundings.

Key ideas to carry forward

- Temperature in kelvin = °C + 273
- $Q = mc\Delta T$ for heating; $Q = mL$ for change of state
- First law: $\Delta U = Q - W$
- Second law: heat flows hot to cold; entropy never decreases; $\eta \leq 1 - Tc/Th$

Module 3.9

Modern physics

Around 1900 classical physics hit three walls: the speed of light, the behaviour of atoms, and radioactivity. The ideas that broke through, relativity, quantum theory and nuclear physics, still shape our technology and our picture of reality.

By the end of this module you can

- Describe the key ideas of special relativity, including time dilation and $E = mc^2$
- Explain photons, wave–particle duality and energy levels
- Describe radioactive decay, half-life, fission and fusion

Special relativity

Einstein started in 1905 from two assumptions: the laws of physics are the same for everyone moving at constant velocity, and the speed of light is the same for every observer. The consequences are strange. Moving clocks run slow, by the factor gamma [Equation 3.9-1]; moving objects shrink along their direction of motion. These effects are tiny at everyday speeds but real: GPS satellites must correct for them or drift by kilometres a day.

The most famous consequence is that mass and energy are equivalent [Equation 3.9-2]. A small amount of mass corresponds to a vast amount of energy, because c squared is enormous. This is the energy source of the Sun and of nuclear power.

Quantum ideas

Light comes in packets called photons, each with energy proportional to its frequency [Equation 3.9-3]. This explained the photoelectric effect, where light knocks electrons out of metals only if its frequency is high enough, regardless of brightness. Einstein won his Nobel Prize for this, not for relativity.

Particles also behave like waves. Electrons fired through two slits make an interference pattern. In atoms, electrons can only occupy certain energy levels; when one drops to a lower level, it emits a photon whose energy equals the difference [Equation 3.9-4]. Each element therefore has its own set of spectral lines, a fingerprint that lets astronomers read the composition of stars, as you will see in Course 5.

The nucleus

Atoms have a tiny dense nucleus of protons and neutrons. Some nuclei are unstable and decay, emitting alpha particles, beta particles or gamma rays. Decay is random for each nucleus but predictable for large numbers: the half-life is the time for half of any sample to decay [Equation 3.9-5]. Carbon-fourteen's half-life of five thousand seven hundred and thirty years makes radiocarbon dating possible.

Heavy nuclei like uranium can split, which is fission, and light nuclei like hydrogen can join, which is fusion. In both, the products have slightly less mass than the starting nuclei, and the missing mass appears as energy through E equals m c squared. Fission powers nuclear reactors; fusion powers the stars.

Key ideas to carry forward

- Light speed is the same for all observers; moving clocks run slow
- $E = mc^2$: mass is concentrated energy
- Photons: $E = hf$; atoms emit specific frequencies
- Half-life is constant; fission and fusion release mass as energy